

USE OF FLORAL DIP-BASED GENE EDITING TO INDUCE RESISTANCE OF POPULAR INDICA RICE VARIETIES TO THE RICE TUNGRO SPHERICAL VIRUS

INTRODUCTION

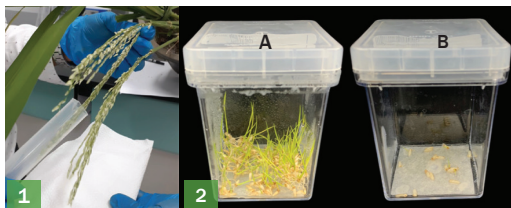
Gene editing, a precise tool, can improve rice crops' traits, such as resistance to rice tungro spherical virus (RTSV) that cause an annual yield loss of 30%. However, traditional plant gene editing technique that largely relies on tissue culture for genetic transformation is very challenging to use for Indica rice varieties owing to their inability to respond to tissue culture manipulations.

The use of floral dip, a non-tissue culture-based gene editing technique, was explored for the first time to genetically transform the pre-anthesis flowers of rice through direct contact with a modified *Agrobacterium tumefaciens*. This bacterium produces a CRISPR-Cas9 (a specific gene editing tool) designed to target the *elf4g* gene to induce resistance to RTSV. Resulting T1 seeds were also screened for resistance to hygromycin (antibiotic); and for genotypic analysis to successfully identify edited lines.

KEY RESULTS

After floral dip, out of the 512 harvested putatively edited T1 seeds from NSIC Rc 402, 422 seedlings turned out to be resistant to hygromycin, demonstrating a high transformation efficiency of 82.42%.

Among the selected hygromycin-resistant T1 plants, 7 exhibited tungro resistance for three consecutive generations, while the wild-type NSIC Rc 402 variety remained susceptible to tungro.



Floral dip experiment: 1. Dipping exposed flowers in *Agrobacterium* suspension; 2. NSIC Rc 402 derived T1 seeds (A) and wild-type NSIC Rc 402 (B) with hygromycin (50mg/L).

IMPLICATION

- Like NSIC Rc 402, a wider range of Indica rice varieties could have a resistance to RTSV through the use of floral dip-based gene editing.
- This technique provides a faster, more cost-effective, and less labor-intensive alternative to traditional tissue culture-based gene editing.

POTENTIAL APPLICATION

- Rice breeding programs can potentially use this technique to rapidly develop improved Indica rice varieties to target other genes for traits, such as yield, grain quality, and resistance to abiotic stress, which are necessary for the attainment of food security and sustainable global agriculture.
- This skips tissue culture, which saves a lot of time while reducing the risk of unintended genome-wide mutations.

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ABOUT THE MATERIAL

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